UbiComp ISWC 2026

Hanyang Human Centered Computing Lab

CLAPS: A Multi-LLM Framework for Predicting Social Anxiety via Cognitive-Behavioral Theory of Social Anxiety in VR

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Motivation

Social Anxiety Disorder → **Exposure therapy** → **Virtual Reality Exposure Therapy**

Severe anxiety in social situations involving potential evaluation by others.

Reduces avoidance behavior by systematically exposing individuals to anxiety-inducing stimuli.

A promising alternative to overcome the temporal and spatial limitations of exposure therapy.

Motivation

How is anxiety measured in VRET?

Post-exposure self-reports

Questionnaires such as PEPQ-R and STAI

Limitations

Cognitive and psychological burdens
Retrospective bias in reporting experienced anxiety

Motivation

How is anxiety measured in VRET?

Post-exposure self-reports

Questionnaires such as PEPQ-R and STAI

→

Limitations

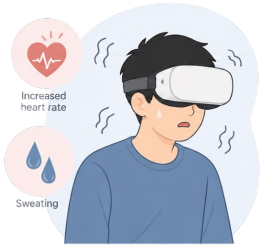
Cognitive and psychological burdens
Retrospective bias in reporting experienced anxiety

Need for objective and automated methods to assess anxiety within VRET environments

Motivation

Types of social anxiety symptoms

Physiological



Behavioral



Cognitive



Physiological and behavioral symptoms captured from sensor signals

Difficult to observe externally or quantify via wearable sensor devices

Motivation

Types of social anxiety symptoms

Physiological **Behavioral**



Physiological and behavioral symptoms captured from sensor signals

Existing prediction models

- Existing prediction models have mainly focused on physiological and behavioral symptoms.
- They predict anxiety using machine learning models (e.g., XGBoost, LightGBM) and deep learning models (e.g., GRU, LSTM).

Motivation

Types of social anxiety symptoms

Modeling cognitive symptoms with LLMs

- Internal cognitive symptoms remain underexplored in data modeling.
- LLMs offer new opportunities to model cognitive symptoms using situational contexts and personal characteristics.

Cognitive



Difficult to observe externally or quantify via wearable sensor devices

Motivation

Why LLMs for cognitive symptoms?

Contextual understanding

Understand complex situational contexts and infer conceptual meanings from text.

Personalized psychological inference

Derive a person's psychological state from their characteristics and situation.

LLMs in mental health research

NarrativeSense

Interprets behavioral context from smartphone logs with LLMs for mental health assessment.

Health-LLM

Uses individual attributes (e.g., emotion, sleep) and clinical knowledge to predict mental health.

→ LLMs open new opportunities to capture cognitive symptoms that sensors cannot directly observe.

Our Approach 1

Theory-based cognitive symptom generation

We use a multi-LLM architecture grounded in the cognitive-behavioral theory of social anxiety to generate scene-level cognitive symptoms.



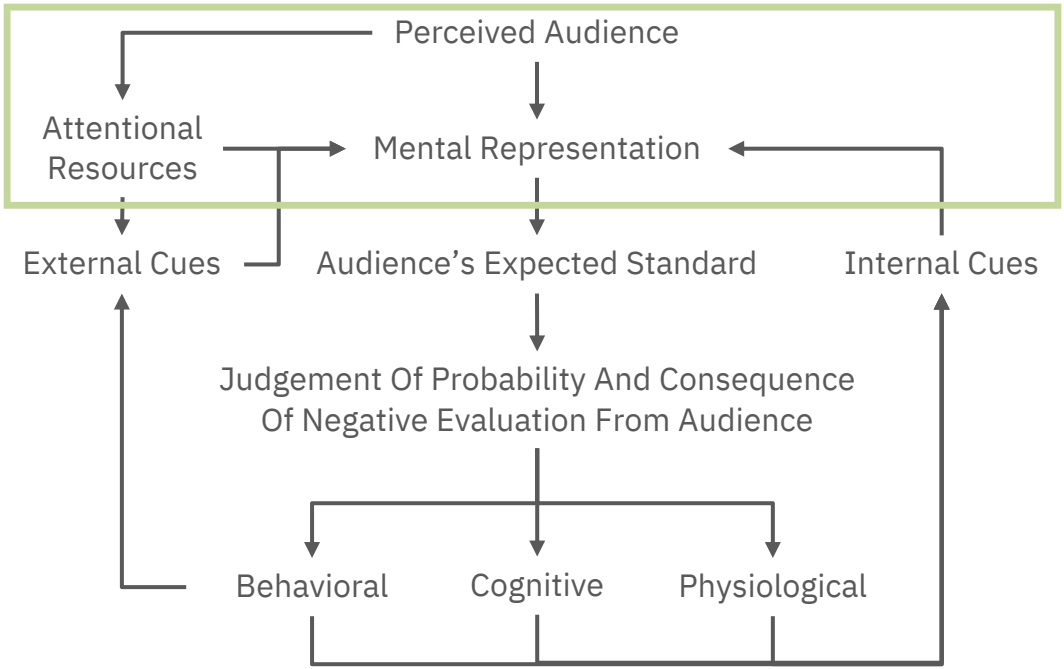
Our Approach 2

Multimodal alignment for anxiety prediction

We align time-series-based physiological and behavioral symptoms with cognitive symptoms at both scenario and scene levels to predict PEPQ-R scores.

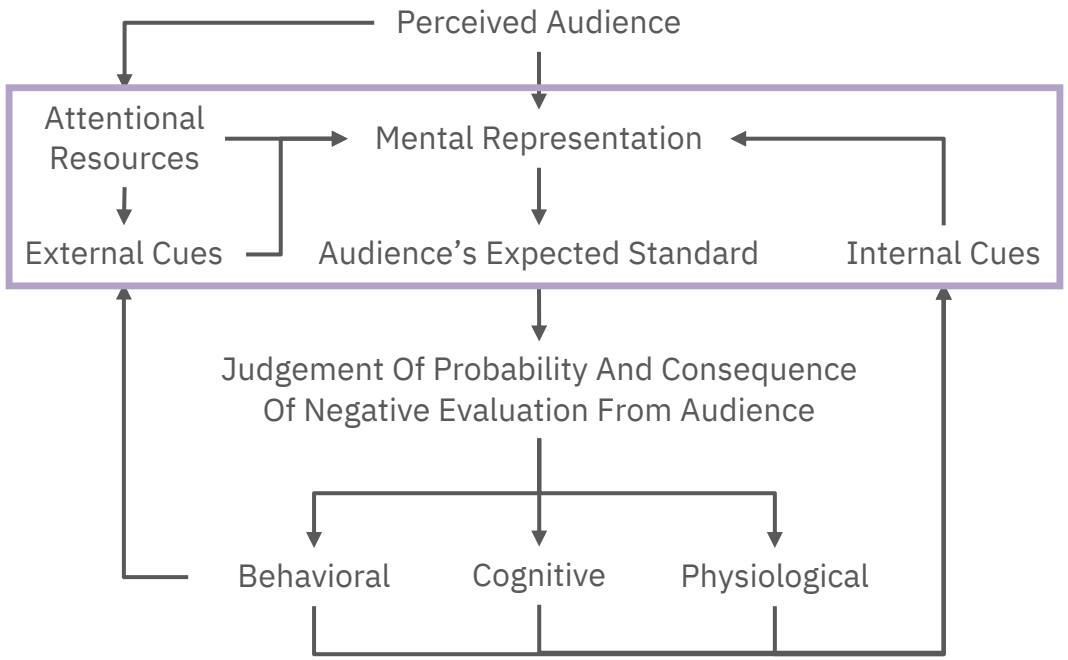


Cognitive-behavioral theory of social anxiety



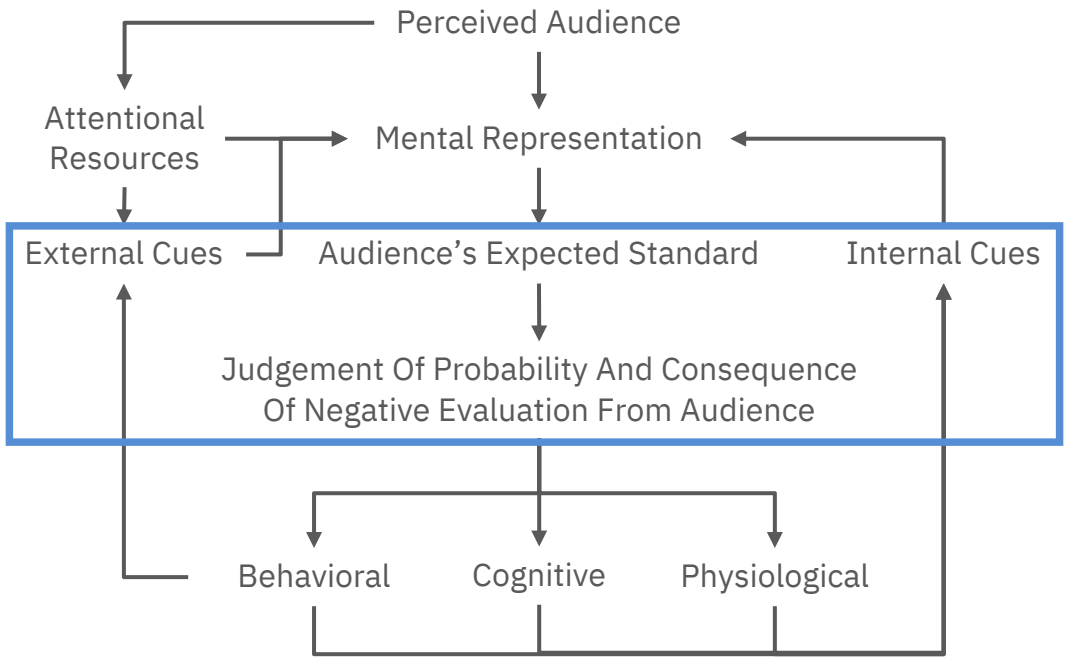
- When a person perceives an audience, long-term memory and the current social context shape the mental representation of self and attentional focus.

Cognitive-behavioral theory of social anxiety



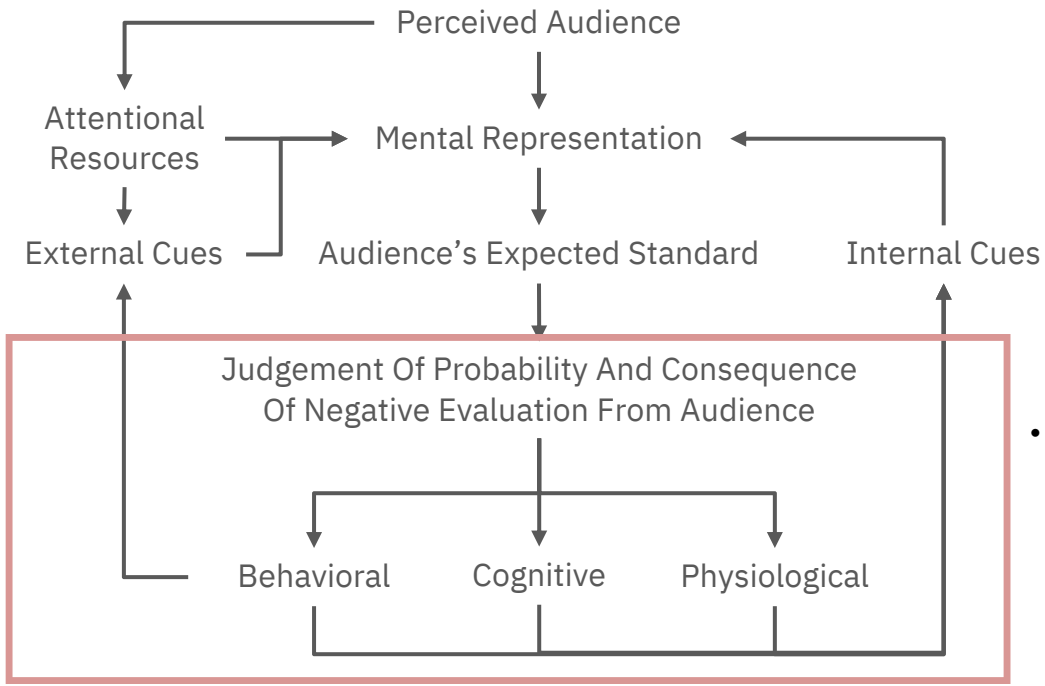
- As the situation unfolds, external and internal cues update the mental representation and the audience's expected standard.

Cognitive-behavioral theory of social anxiety



- When the mental representation is compared with the audience's expected standard, the person judges the probability and consequences of negative evaluation.

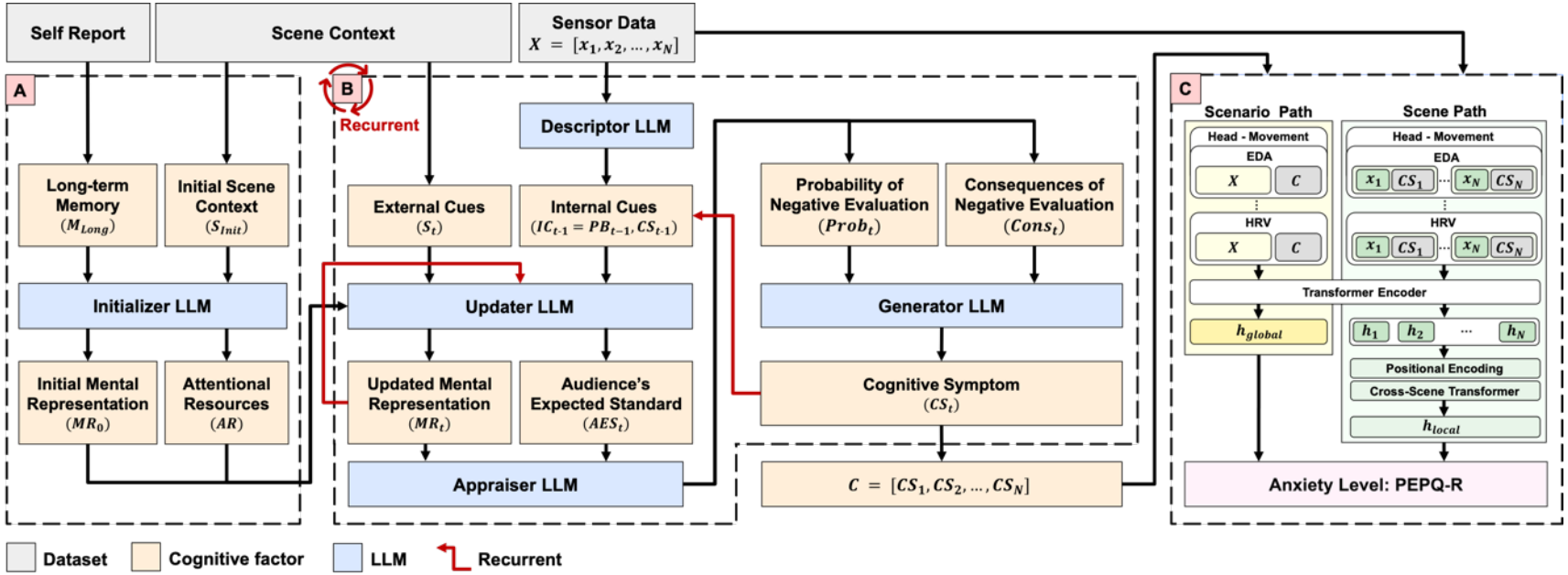
Cognitive-behavioral theory of social anxiety



- When negative evaluation is expected, physiological, behavioral, and cognitive symptoms occur and feed back into the cycle.

Framework Overview

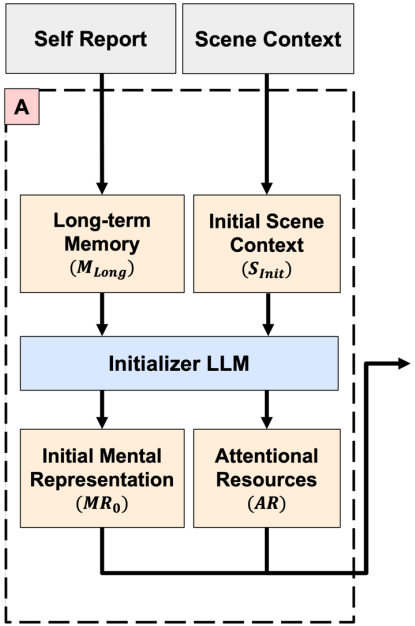
CLAPS (**C**ognitive-behavioral **L**LM-**A**ssisted **P**rediction of **S**ocial anxiety)





Initialization

Establishing the initial cognitive factors



Inputs

- Long-term memory: pre-session self-reports
- Initial scene context

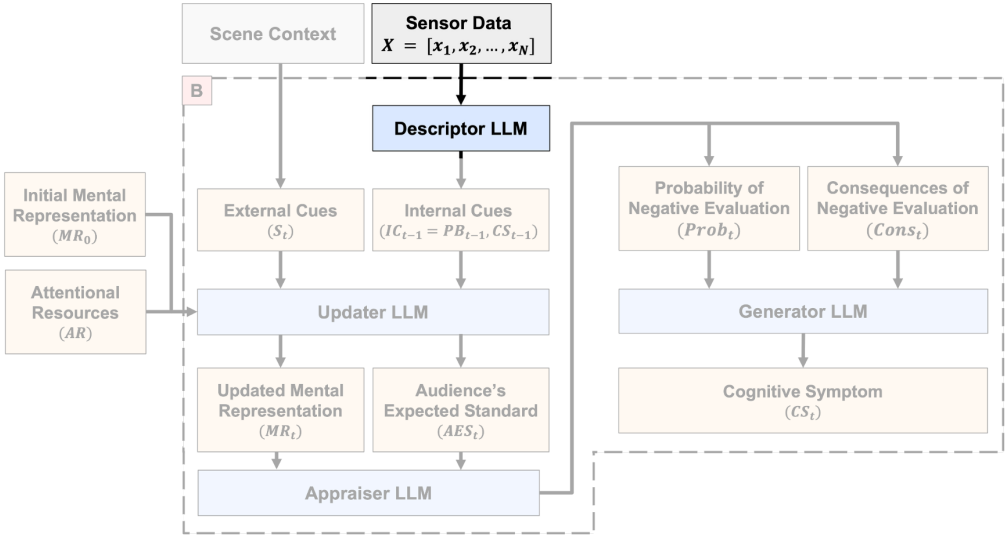
Initializer LLM

- Generates the initial mental representation (MR_0)
- Generates the preferential allocation of attentional resources (AR)



Cognitive Symptom Generation

From cognitive factors to scene-level cognitive symptoms



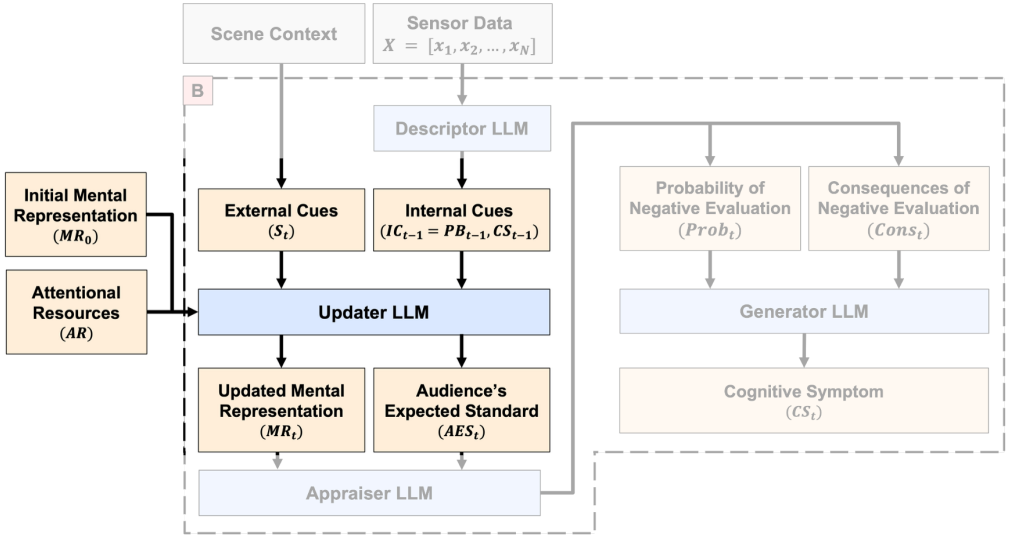
Descriptor LLM

- Describes physiological and behavioral time-series changes as text (PB_t).



Cognitive Symptom Generation

From cognitive factors to scene-level cognitive symptoms



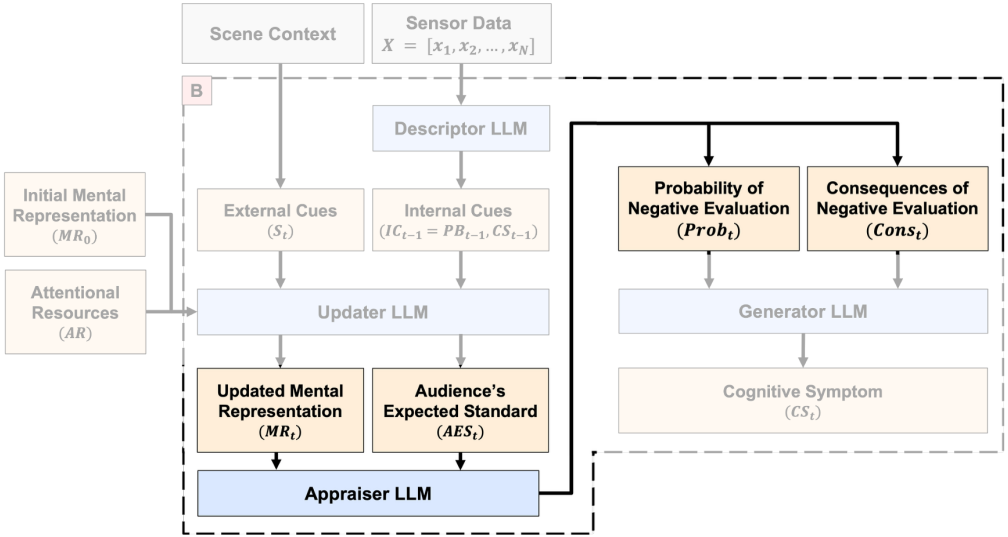
Updater LLM

- Updates the mental representation (MR_t)
- Updates the appraisal of the audience's expected standard (AES_t).



Cognitive Symptom Generation

From cognitive factors to scene-level cognitive symptoms



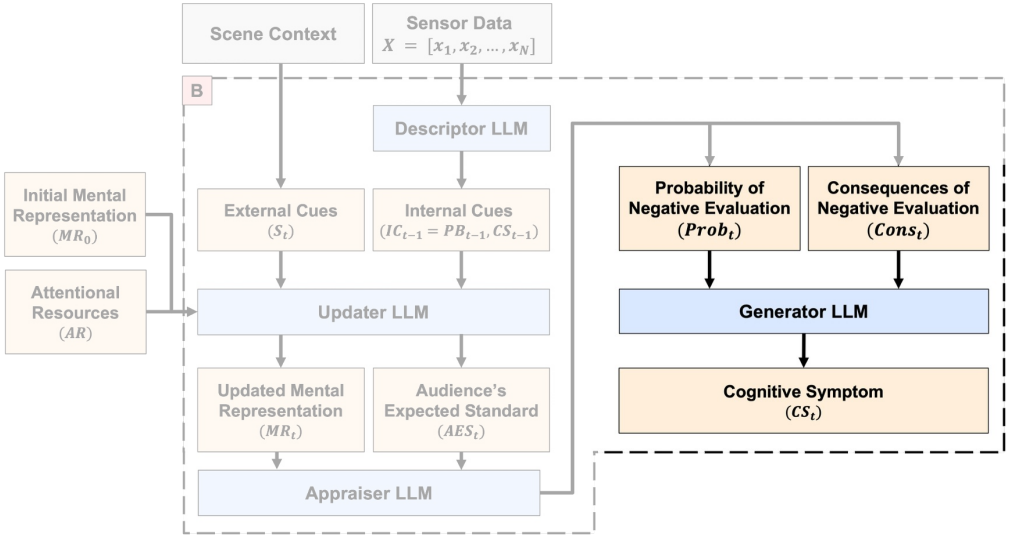
Appraiser LLM

- Infers the probability ($Prob_t$) and consequences ($Cons_t$) of negative evaluation.



Cognitive Symptom Generation

From cognitive factors to scene-level cognitive symptoms



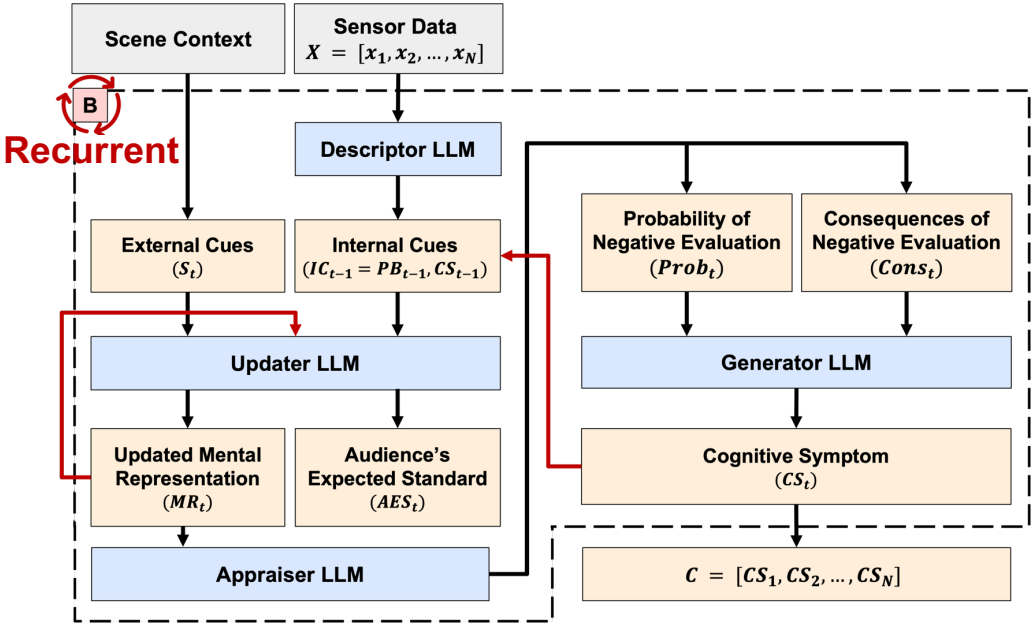
Generator LLM

- Generates a scene-level cognitive symptom (CS_t) from these appraisals.



Recurrent

Using previous-scene symptoms to inform the next scene



Recurrent Structure

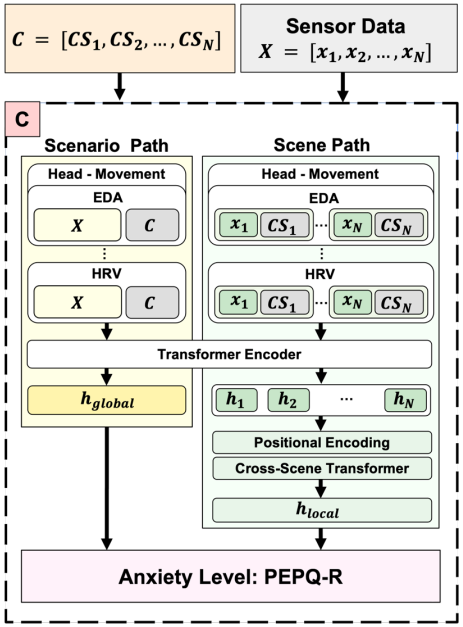
- For each scene, the Updater LLM uses the previous mental representation and internal cue.
- The internal cue contains the previous scene's sensor description (PB_{t-1}) and cognitive symptom (CS_{t-1}).

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Cross-modal Aligned Transformer

Aligning symptoms at scene and scenario levels



Inputs

- Physiological and behavioral time-series data (X).
- Generated cognitive symptom sequence (C).

Scenario Path

- embeds the X and C to generate a global embedding (h_{global}).

Scene Path

- Aligns x_t and CS_t from the same scene to generate h_t.
- Uses positional encoding and the Cross-Scene Transformer to generate a local embedding (h_{local}).

Prediction

- Combines global and local embeddings to predict nine PEPQ-R item scores.

Dataset

2 VR scenarios



Self-introduction



Job interview

32 Participants

30 Time-series features

- EDA
- HRV, BVP
- Eye-tracking
- Head-movement

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2 VR scenarios



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Data Collection Process

Pre-survey

Self-Presentation Motivation
Self-Presentation Expectancy

During each scenario

Collecting physiological and behavioral time series data

Post-survey

Nine PEPQ-R items
7-point Likert scale

Experiment

CLAPS achieved best score among the evaluated models

Two-sided paired Wilcoxon signed-rank test / * p < 0.05; ** p < 0.01; *** p < 0.001

Model	MAE	RMSE
Random Forest	1.745 ± (0.278)***	2.300 ± (0.270)***
XGBoost	1.703 ± (0.201)***	2.235 ± (0.221)***
LightGBM	1.731 ± (0.235)***	2.228 ± (0.242)***
SAFER [9]	1.459 ± (0.271)**	1.769 ± (0.274)**
DepGuard [29]	1.384 ± (0.218)	1.728 ± (0.227)**
Health-LLM [13]	1.564 ± (0.327)***	1.995 ± (0.336)***
CLAPS w/o Cognitive Symptom	1.369 ± (0.223)	1.664 ± (0.271)
CLAPS w/o Scene Path	1.344 ± (0.172)**	1.619 ± (0.201)**
CLAPS (ours)	**1.268 ± (0.180)**	**1.558 ± (0.183)**

Value of cognitive symptoms

- cognitive symptoms provide additional information for anxiety prediction.
- LLMs show potential to generate internal states from contextual cues.

Experiment

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Importance of scene-level alignment

- Aligning physiological, behavioral, and cognitive symptoms improves prediction.
- Both scenario-level and scene-level temporal patterns should be considered.

Limitation and Future Work

Data requirements

Requires self-reports and structured social context descriptions.

↓

Estimate personal factors from sensor data, and generate social context descriptions with VLMs.

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Theory-specific

The architecture is tailored to the cognitive-behavioral theory of social anxiety.

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Build a modular framework for different cognitive-behavioral theories.

Limitation and Future Work

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Theory-specific

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Build a modular framework for different cognitive-behavioral theories.

Cognition validity

Generated symptoms may not reflect users' actual cognitive experiences.

↓

Validate generated symptoms through expert reviews and user studies.

Conclusion

01 **Theory-based cognitive symptom generation**

CLAPS generates cognitive symptoms grounded in the cognitive-behavioral theory of social anxiety.

02 **Multimodal alignment**

CLAPS aligns generated cognitive symptoms with co-occurring physiological and behavioral data.

03 **Anxiety prediction**

CLAPS outperformed all baseline models, achieving an MAE of 1.268 and an RMSE of 1.558.

Thank you!

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Paper